

Symbolic Material Culture in the Late Pleistocene: Use in Prehistory, Appearance in the Archaeological Record and Taphonomy

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INTRODUCTION

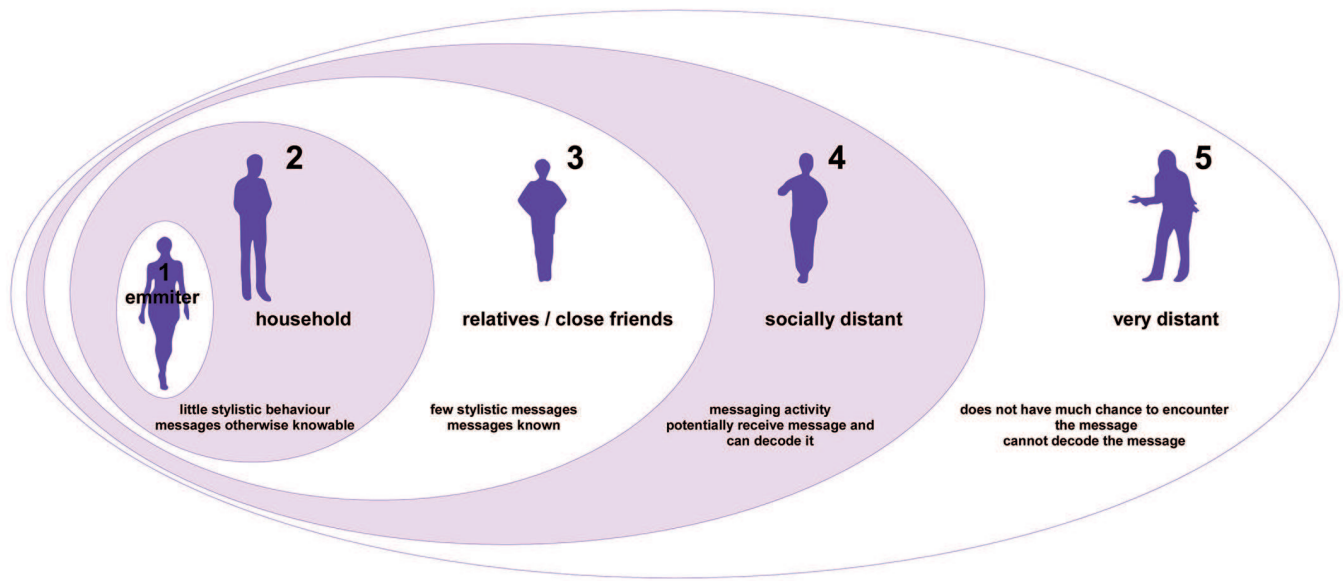
People all over the world use material culture to transmit information about themselves to observers every day. This was no exception in the past. At some point in human prehistory, either owing to changes in the environment, population densities or some other cause, the cognition and behaviour of Modern Human populations in Africa, and perhaps Neanderthals in Eurasia, developed and changed so that artefacts were invested with abstract meanings and symbolic values for the first time in human prehistory (Mithen 1996), rather than being manufactured solely for utilitarian purposes (Bisson 2001). When this ability to use material culture to transmit information developed in human evolution is a topic that has been heatedly debated (e.g. Bouzouggar, Barton, Vanhaeren et al. 2007, Brumm, Moore 2005, Conard 2008, d’Errico 2003, Henshilwood, Marean 2003, Henshilwood, d’Errico, Vanhaeren et al. 2004, Henshilwood 2007, Klein 2000, Mellars 2005, Wadley 2001).

Through focusing on four specific material culture categories – body ornamentation, hunting weaponry, mobiliary art and rock art – this paper will not only outline how symbolic material culture may have been used during the late Pleistocene by human communities but also highlight the problems associated in interpreting their presence (and absence) from the archaeological record.

USE IN PREHISTORY

Symbolic behaviour is “*the ability to represent objects, people, and abstract concepts with arbitrary symbols, vocal or visual, and to reify such symbols in cultural practice*” (McBrearty, Brooks 2000: 492). A key characteristic of all symbol use is that their meaning is assigned by arbitrary, socially-constructed conventions (Chase, Dibble 1987), allowing the storage and display of information external to the human brain (Donald 1991, Henshilwood, Marean 2003, Hovers, Ilani, Bar-Yosef et al. 2003, Wadley 2001).

Artefacts which are commonly believed to represent symbolic behaviour in the archaeological record include: regional artefact styles, body ornamentation such as beads and pigments or the modification of the body itself (e.g. cultural cranial deformation, piercings, tattoos), notched and incised items (bone, egg shell, ochre, stone, antler etc), ritual burials, mobiliary art and rock art (Ambrose 1998, Chase, Dibble 1990, Deacon 2001, d’Errico, Henshilwood, Vanhaeren et al. 2005, Klein 1995, McBrearty, Brooks 2000, Mellars 1989a, 1989b, 2005, Renfrew 1996, Thackeray 1992). These artefacts and features when found in the archaeological record reflect not only the ability of



a population to use abstract concepts, but also the use of these concepts to link the individual or population to their technologies and environment (Henshilwood, Marean 2003).

Another aspect of symbolic material culture, and arguably the greatest benefit of symbolically mediated behaviour in general, is its ability to link the individual or group to other individuals and groups by transmitting information between populations. This ability for material culture to transmit information to observers is what Wobst (1977) has termed 'artefacts as information exchange' (also see Weissner 1983, Gamble 1980, 1982). Using artefacts to broadcast information has its primary benefit in that the emitter of the information (the person who made and/or owns the artefact) does not need to be present for the message to be transferred to the observer. Unlike vocal messages where the emitter and the receiver must be within 'earshot' and last only as long as the sound carries, messages held in artefacts have relative longevity, do not change rapidly, and can be made portable and therefore broadcast their message widely around the landscape and to people who are distant to the emitter (Wobst 1977).

The use of artefacts to convey messages are most useful (valuable) when used to communicate information to those people in 'the middle distance'. That is, those people not so close to the emitter that the messages are already known (groups 2 and 3 in Figure 1) and not so distant that the meaning of the message cannot be deciphered (group 5 in Figure 1). This 'middle group' (group 4 in Figure 1) – those people who are not encountered everyday but frequently enough that they understand the meaning of displayed symbols, and may include neighbouring groups, trading partners, or other alliance members – is where the majority of symbolic artefacts and stylistic behaviour is aimed.

As Wobst (1977) explains, because stylistic messages are primarily aimed at 'the middle distance', these messages will mostly concern social integration and social differentiation. Furthermore, as complex messages or messages which are only infrequently emitted require more energy to both make (emit) and interpret (receive), only simple and frequently used messages will normally be encoded in material culture. The types of messages which will therefore be commonly broadcast include: emotional state, identification (class affinity, social group affiliation, religious affiliation, rank, etc.), authorship and ownership.

The use of material culture to transmit messages between the emitter and those in 'the middle distance' has a number of advantages for both parties. These include the reduction of stress in first encounters between groups as the artefacts broadcast the potential advantages and disadvantages of the encounter before it has occurred (Wobst 1977), and acting to bind groups together in times of economic and/or social stress allowing for cooperation between related groups and the continued survival of all parties (Gamble 1982). In Conard's (2008) words, these artefacts acted as the social

Fig. 1
Target groups for
stylistic messages
held in the design
and display of
material culture.

and symbolic 'glue' that held together larger and more complex social groups than those which existed in earlier human prehistory.

Throughout the late Pleistocene messages held in artefacts were carried (transmitted) along pathways forged by alliance networks which mediated the circulation, control and exchange of information as well as goods and people (Bender 1978, 1985). From an information exchange standpoint then, the temporal and spatial patterns of symbolic material culture can be used to examine the volume and diversity of information flow within and between regions during prehistory. This approach is one which many researchers have followed to examine the development and growth of Pleistocene social networks as well as to attempt to identify when a group/s were under environmental and/or social stress (e.g. Bahn 1982, Barton, Clark, Cohen 1994, Conkey 1980, Gamble 1982, Jochim 1983, Vanhaeren, d'Errico 2006).

In order to effectively (and briefly) outline the potential use of symbolic material culture during the late Pleistocene and the impact various anthropogenic and taphonomic processes have had on the creation of the archaeological record, four categories of material culture will now be examined: body ornamentation, hunting weaponry, mobiliary art and rock art. These four particular categories of material culture were chosen as they enable the discussion of the different uses of style and symbols at different levels of communication during prehistory. They also provide excellent examples of how taphonomy can impact on that which we have left to build our interpretations of prehistoric behaviours, a point which will be specifically addressed below.

BODY ORNAMENTATION

Body ornamentation includes any item which is used to enhance or decorate an individual's person. This includes clothing, pigment to be painted onto skin or clothing, beads and ornaments, as well as tattoos or other physical modification to the body itself. The most numerous of these items to be recovered from the Pleistocene archaeological record are beads and pigments.

Beads and ornaments are small, durable objects which have been modified (through perforations and grooves) so that they might be attached or suspended to other materials. These can be displayed against human skin (as in necklaces, earrings, hair pieces etc), attached to clothing or other equipment which are then carried around the landscape by the individual (Kuhn, Stiner 2007).

In almost every ethnographic and contemporary Modern Human society, beads, clothing, pigment, and any other media used to decorate the body are used to convey messages (see for example: Brain 1979, Corwell, Schwarz 1979, Dubin 1987, 2003, Faris 1972, Hodder 1991, Kuper 1973, O'Shea 1984, Preston-Whyte 1994, Roach, Eicher 1965, Sciama, Eicher 1998, Wiessner 1984). These messages transmit information to a variety of audiences both within and outside of the wearer's community (as outlined above), about the wearer's identity. This information can include age, group affiliation, marital status, social standing, personal achievements and wealth and will be determined by the wearer's lineage (clan, moieties), gender, age class (child, adolescent, adult, elder), relational status (single, married, parent, widowed) as well as personal achievements (Kuhn, Stiner, Reese et al. 2001, Kuhn, Stiner 2007).

In the archaeological record, the earliest undisputed beads in Modern Human contexts are perforated *Nassarius gibbosulus* shells from Skhul (Mt Carmel, Israel) found in layers dating to >100,000 yr BP (Vanhaeren, d'Errico, Stringer et al. 2006). These same kinds of ornaments (perforated *Nassarius gibbosulus* shells) have also been recovered from Grotte des Pigeons (Taforalt, Morocco) in levels dated to ~82,000 years ago (Bouzouggar, Barton, Vanhaeren et al. 2007). Shell beads (*Nassarius kraussianus*) have also been recovered from the Middle Stone Age (MSA) levels (~75,000 years) of Blombos Cave in South Africa (Henshilwood, d'Errico, Vanhaeren et al. 2004, d'Errico, Henshilwood, Vanhaeren et al. 2005). The appearance of shell beads and beads manufactured from other raw materials such as teeth, bone, antler and stone become increasingly common and standardised in later Pleistocene contexts

throughout Africa, Eurasia, Asia, and Australia (see for example d’Errico, Henshilwood, Vanhaeren et al. 2005, Kuhn, Stiner 2007, Vanhaeren, d’Errico 2005, White 2007).

Beads and other body ornaments in Neanderthal contexts remain heatedly debated (e.g. d’Errico 2003, d’Errico, Zilhão, Julien et al. 1998, d’Errico, Henshilwood, Lawson et al. 2003, Mellars 2005, Zilhão 2007), however recent discoveries at Cueva de los Aviones and Cueva Antón (Spain) of perforated and pigment-stained marine shells (*Acanthocardia*, *Glycymeris* and *Pecten*) have provided robust examples of the use of this technology by Neanderthal populations around 50,000 years ago (Zilhão, Angelucci, Badal-García et al. 2010). Neanderthals may also have used bird feathers to ornament either themselves or their equipment as evidenced by human modifications on species that are not clearly a food source (including Eurasian black vulture, golden eagle, and red-footed falcon among others) found at Grotta di Fumane, Italy (Peresani, Fiore, Gala et al. 2011). The authors argue that the cut, peel, and scrape marks along with fractures found exclusively on the wings of the animals indicate the intentional removal of large (and often most colourful or distinctive) feathers by Neanderthals at this site around 44,000 years BP.

Unlike beads, mineral pigments may have utilitarian purposes, such as polishing organic artefacts, the preservation of hides, and hafting (Mellars 1996: 370, Minzoni-Deroche, Menuet, Walker 1995, Wadley 2005) as well as being used in a symbolic sphere of behaviour (body painting, painting of artefacts, rock art etc). While beads can be standardised through their selection, the modification of natural forms and the manufacture of replica natural form (such as shells and teeth) beads out of stone, pigment can only be standardized through its colour and its use (Kuhn, Stiner 2007).

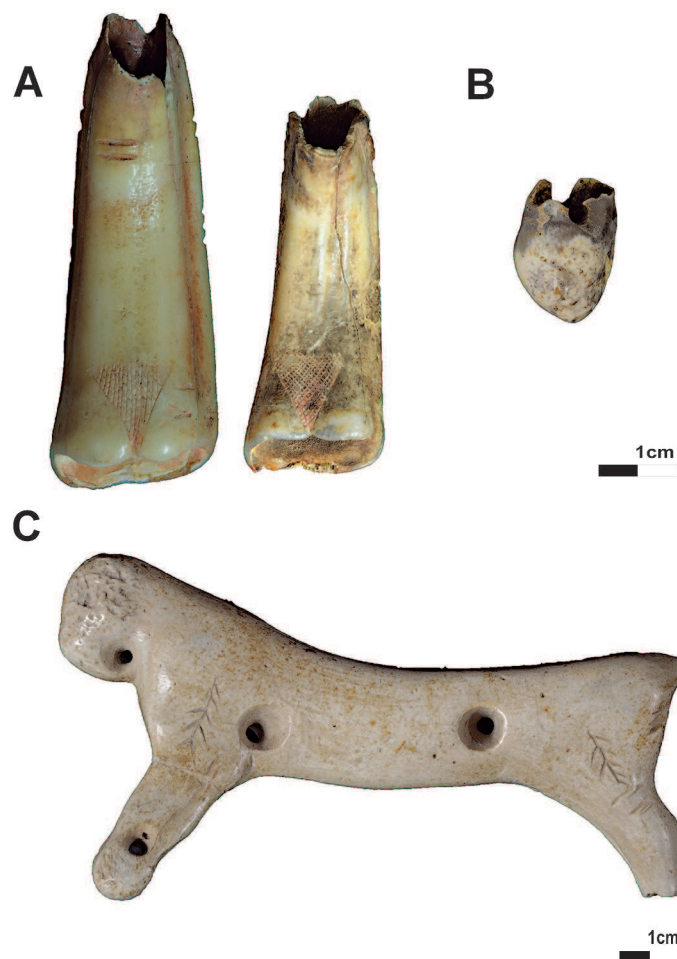


Fig. 2
Examples of Late Pleistocene beads and portable art. (A) Two engraved horse teeth. Left: La Marche (Magdalenian), Right: Lussac (Magdalenian). (B) Deer tooth bead – Grotte de l’Abbe (Magdalenian). (C) Carved antler lion – Isturitz (Magdalenian).

Ochre is only recognised as pigment by archaeologists because it is usually exotic to the site, and because it shows evidence of working (striations and other signs of wear). Palaeolithic sites often also hold other materials (e.g. terra rosa clays, charcoal, white ash) that might also have been used as pigments but leave no evidence for having been used in this way.

The earliest solid evidence for the use of pigments – in the form of worked red ochre as well as grinding implements – currently comes from the Kapthurin formation (Kenya) at <285,000 years (McBrearty 2007, McBrearty, Brooks 2000), with similar finds of ochre dated to ~230,000 years at Twin Rivers Kopje (Zambia) (Barham 1998, Barham, Smart 1996). Ochre, however, is present throughout the MSA and in even earlier sites throughout sub-Saharan Africa though without clear evidence of use (Barham 1998, McBrearty, Brooks 2000). In western Europe, manganese dioxide, red ochre, and other materials with clear signs of use are common in Middle Palaeolithic (Neanderthal) contexts as early as oxygen isotope stage 5 (e.g. Hovers, Ilani, Bar-Yosef et al. 2003, Mellars 1996).

A recent discovery at Blombos Cave (South Africa) has provided an exceptional example of an ‘ochre processing workshop’ dated to ~100,000 years ago (Henshilwood, d’Errico, van Niekerk et al. 2011). Found *in situ*, the tool kit includes a “liquefied ochre-rich mixture” stored in two abalone (*Haliotis midae*) shells, along with bone, charcoal, grindstones and hammerstones. While what the ochre mixture was applied to is unknown, and could include both utilitarian (hafting, skin preparation, etc) or symbolic (body painting, decoration of artefacts, etc) tasks, this find provides a rare insight into how pigment was prepared and stored during early Modern Human prehistory.

Scarification, piercing and tattoos might also have been undertaken during the late Pleistocene, though no evidence for this behaviour survives – nor could – to present day. The earliest evidence for tattoos is the (in)famous ‘iceman’ discovery, dating back to 5,300 years and which survived to present day through fortuitous preservation circumstances (Pabst, Letofsky-Papst, Bock et al. 2009). Cranial deformation on the other hand, can and does survive in some circumstances. Some of the most impressive examples of Pleistocene cultural cranial deformation come from Australia – in particular, the Kow Swamp 5 crania dated to c. 13,000 years BP, the Cohuna crania dated to c. 11,000 years BP and the Coobool Creek 65 crania dated to c. 12,500 years BP (Antón, Weinstein 1999, Brown 1981). This practice may also have been undertaken by Neanderthals in Western Asia as suggested for two individuals (1 and 5) recovered from Shanidar (Iraq) and dated to ~46,000 years BP (Ivanhoe 1983, Trinkaus 1982, 1983).

While it is impossible to determine what particular messages were being transmitted by the decoration worn on an individual’s body, we can talk about how, where, when and how much such decorative technologies were used during the late Pleistocene. However, with the exception of a few exceptional cases of preservation in which hundreds of beads and other ornaments were recovered (e.g. the Sungir burials; Bader 1970, Formicola, Buzhilova 2004), archaeologists only have single, isolated beads or very small assemblages of them along with the discarded remains of used pigments. Clearly, what has survived to present day is only the ‘tip of the iceberg’ and we must remember this fact when investigating how these items were used in the late Pleistocene.

What is known, however, is that Pleistocene peoples appear to have restricted their selection of natural materials (shells, teeth, claws, etc) to only a few specific types which were then modified in particular ways creating a widespread standardisation of ornament use. As has been often noted, the beads from Blombos Cave, Grotte des Pigeons, Skhul and other sites of similar (and later) ages are all of a single species (i.e. *Nassarius gibbosulus*, *Nassarius kraussianus*) (e.g. Kuhn, Stiner 2007, Stiner 2003, Taborin 1993). Ornament assemblages show similarly restricted selection of natural forms (animal teeth in particular) or manufactured designs, such as basket-shaped ivory beads (White 1992, 1993) or ostrich-eggshell disks, with some manufactured forms made to resemble natural forms (particularly shells and teeth) (White 1989, 1993). The use of beads and ornaments for adornment, therefore, while first appearing in the Middle Palaeolithic and MSA, “*codified use of discrete objects as ornaments seems to associate specifically with the appearance of the EUP [early Upper Palaeolithic] and late MSA*” (Kuhn, Stiner 2007: 47).

MOBILIARY ART

Portable, or mobiliary, art while often including the beads and ornaments discussed above, will here refer only to small pieces of sculpture, engravings or paintings executed on stone, bone, ivory, antler or any other raw material which is relatively small in size and can therefore be easily transported around the landscape.

The earliest examples of this kind of artefact to be found in Africa are painted granite slabs found at Nswatugi (Zimbabwe) in late MSA levels (>42,000 years BP) and from Apollo 11 cave (Namibia) also in a MSA horizon. These pieces may, in fact, be painted roof fall, however, Wendt (1976) argues that they were portable art. Notched pieces of bones have also been recovered from a wider range of African Middle and Late Stone Age sites, including Klasies River (South Africa), Apollo 11 (Namibia), Border Cave (Swaziland) and Blombos Cave (South Africa) (Beaumont 1978, Henshilwood, Sealey 1997, Singer, Wymer 1982, Vogelsang 1998, Wendt 1972, 1976). Similarly, incised pieces of eggshell which may or may not have been pieces of water containers have been found at Diepkloof (South Africa) and Apollo 11 (Namibia) (Texier, Porraz, Parkington et al. 2010, Wendt 1972, Vogelsang 1998). Finally, the incised ochre pieces recovered from Blombos Cave must be mentioned (Henshilwood, d'Errico, Watts 2009). Here, two deliberately engraved pieces of ochre from c.75,000 year old levels were recovered – each with a cross-hatched pattern constituting the oldest evidence for abstract design in a Modern Human context.

The Pleistocene Australian archaeological record shows a similar pattern to that of Africa in regards to mobiliary art. Here only four artefacts have been thus far identified which clearly fall into this category of symbolic material culture. These consist of two engraved limestone plaques dated to 24,531 cal BP and 13,815 cal BP respectively from Devil's Lair, Western Australia (Dortch 1976). One of these displays a trapezoidal shape while the other has no clear design.

The two other instances consist of incised bone. The first is a macropod femur with incised grooves dated to 21,803 cal BP from Cave Bay Cave, Hunter Island (Stockton 1981) and the other was found at Spring Creek, Victoria dated to 32,602 cal BP and is a *Diprotodon optatum* incisor with 28 incised grooves. This last instance is controversial owing to the fact that if these incisions are anthropogenic as is suggested by Vanderwal and Fullagar's (1989) analysis, and the incisor was not simply picked up off the landscape, then this may be evidence for interaction between humans and the Australian megafauna.

Eurasia provides a significantly richer array of portable artworks. The Neanderthal archaeological record has provided a number of cases of notched and incised bones and stones suggested to be the product of intentional marking, the majority of these, however, have been dismissed as the result of microscopic analysis of the pieces (e.g. d'Errico, Villa 1997). A handful of artefacts still remain, however, and include a fossil nummulite with a cross incised (Tata, Hungary c.100,000 years) (Marshack 1990, 1996), notched and incised bones (Bacho Kiro, Bulgaria c. 44,000 years; Tagliente Rockshelter, Italy between 30,000 and 60,000 years BP) (Fiore, Gala, Tagliacozzo 2004, Leonardi 1983, 1988, Marshack 1976, 1996, d'Errico, Villa 1997), engraved schist and other stone plaques (Tagliente Rockshelter, Italy between 30,000 and 60,000 years BP; Temnata Cave, Bulgaria, c.50,000 years BP (Crémades, Laville, Sirakov et al. 1995, d'Errico, Villa 1997, Fiore, Gala, Tagliacozzo 2004, Leonardi 1983, 1988) and a carved and polished section of mammoth dental plate with ochre stains found at Tata (Hungary) dated to ~50,000 years (Marshack 1976).

When Modern Humans arrived in Eurasia they either brought with them or quickly developed a strong tradition of creating small portable artworks. Engravings of animals, humans and abstract designs on small pieces of antler, bone, or stone have been found dating from the Aurignacian and continuing right through to the late Magdalenian at the close of the Pleistocene (see Graziosi 1960 for examples from each period). While some suggest a religious focus for these items – usually referred to as plaquettes (e.g. Leroi-Gourhan 1965, Sieveking 1987, 1991), there seems to be some consensus that it was the *act of creating* the engraving which was important. Having been made, these items were then discarded presumably having no further role to play in Palaeolithic (religious) life

after the event in which they were manufactured was concluded. These plaquettes, unlike parietal art which survived in a relatively protected environment, would then be exposed to trampling and other types of damage commonly accrued through exposure to everyday activities due to being discarded on a shelter (or other site) floor. Additionally, they were often reused and recycled in domestic activities (such as lining a hearth or a shelter floor) suggesting that their use life continued in possibly an entirely different function to that which it was originally created for (Sieveking 1987).

Small sculptures of animals, humans (particularly women) and therianthropes (animal – human combinations) also appear early in the Aurignacian indicating that they were part of a much older tradition of art (Conard 2009) and continue throughout the Upper Palaeolithic of Europe. While the meaning of these figurines is unknown, theories include hunting magic (Reik 1934), shamanism (Lewis-Williams 2002) and, in the case of the famous ‘venus’ figurines, fertility (Delporte 1993).

Owing to the portable nature of these items, the actions which led to their manufacture, use and discard as well as any potential spatial patterning which may have been present within their use lives is potentially erased by their portability. Certainly there is evidence from a number of late Pleistocene examples, such as the engraved rib with a horse head and forelimb behind a series of lines from Robin Hood’s Cave and the ivory horse from Lourdes which show a high degree of handling polish suggesting that these smaller artworks may have been carried extensively around the landscape and perhaps handed down from one generation to the next (Sieveking 1991, 1992).

Despite these uncertainties, however, some researchers have concluded that mobiliary art pieces “*were done in the context of ordinary life, and were directed at smaller domestic, social audiences*” (Rice and Paterson 1996: 232) as opposed to the wider ‘middle distance’ audience that body ornamentation was primarily aimed at. Gamble (1982) along with Barton, Clark, Cohen (1994) argue that these items additionally functioned at the wider regional level of interaction as, in the case of the Venus figurines, they appear at a time when environmental conditions were deteriorating and open social networks were to be expected. The stylistic similarity of these objects perhaps provide an indication of the strain being placed upon alliances in the mating network and contributed to keeping groups connected during poor conditions so that they could continue to subsist in difficult environments.

ROCK ART

Unlike mobiliary art, parietal art was probably created to transmit a clear message to a defined target population (Barton, Clark, Cohen 1994), and would have acted to facilitate inter-group communication, messaging across social and territorial boundaries.

Under conditions of low population density, knowledge of landscape and people is primarily transmitted from one generation to the next by means of oral tradition; the intentional marking of the landscape was sparsely required. As population density increases, however, the need for more visual based cues in the form of distinctive material culture and rock art to demarcate significant localities became necessary. As population density increases regionally, further marking of the landscape to demarcate boundaries and legitimise group identities and claims to resources (that is, increased territoriality), and, in particular, to communicate these claims to other social groups is required (Barton, Clark, Cohen 1994, Conkey 1978, 1980, Gamble 1982, Hodder 1979, Jochim 1983, Taçon 1994, White 1985, 1982, Zedeño 2008).

This use of rock art to demarcate territories, identify resident groups and maintain social cohesion is recognised in various ethnographic contexts (e.g. Denbow 1984, Lee, Hyder 1991, Lewis-Williams 1984, McDonald 2008, Taçon 1994) and has been suggested to include various late Pleistocene culture areas including the European Palaeolithic (Barton, Clark, Cohen 1994, Jochim 1983, Straus 1982, 1987).

The antiquity of rock art in Sahul is vigorously debated (e.g. Gillespie 1997, Langley, Taçon 2010, Loy 1994, Loy, Jones, Nelson et al. 1990, Nelson 1993, Rosenfeld, Horton, Winter 1981, Trezise 1992).

The oldest evidence for rock art in Sahul comes from Carpenter's Gap Rockshelter 1, Western Australia and consists of a piece of painted roof fall recovered from the sites Pleistocene deposit, dated securely to 42,800 – 38,177 years BP (O'Connor, Fankhauser 2001). In Africa, painted and engraved outcrops of rock and caves date back to the LSA and have a wide distribution throughout sub-Saharan Africa (McBrearty and Brooks 2000).

Rock art in Europe begins, as with mobiliary art, with the arrival of Modern Humans into the region who produced the extraordinary art in Chauvet Cave (Ardèche, France) dating as far back as 32,000 years (Clottes 2001, Pettitt, Bahn 2003) and the numerous later Magdalenian painted caves – including Lascaux and Niaux – found throughout Western Europe.

Hunting Weaponry

The Upper Palaeolithic of Europe is famous for its extensive array of engraved and decorated osseous (antler, bone, ivory, horn) weapons. This is particularly so for the Magdalenian, where significant amounts of time and effort were put into decorating projectile weaponry such as spear-throwers, barbed points (*harpons*) and unbarbed points (*sagaie*) (Figure 3). Diagnostic lithic projectile weaponry was also produced by Upper Palaeolithic Europeans which, like osseous weaponry, is argued to have been imbued with messages through standardised form (e.g. Dibble 1989, Mellars 1989b, see Wiessner 1983 for an ethnographic example).

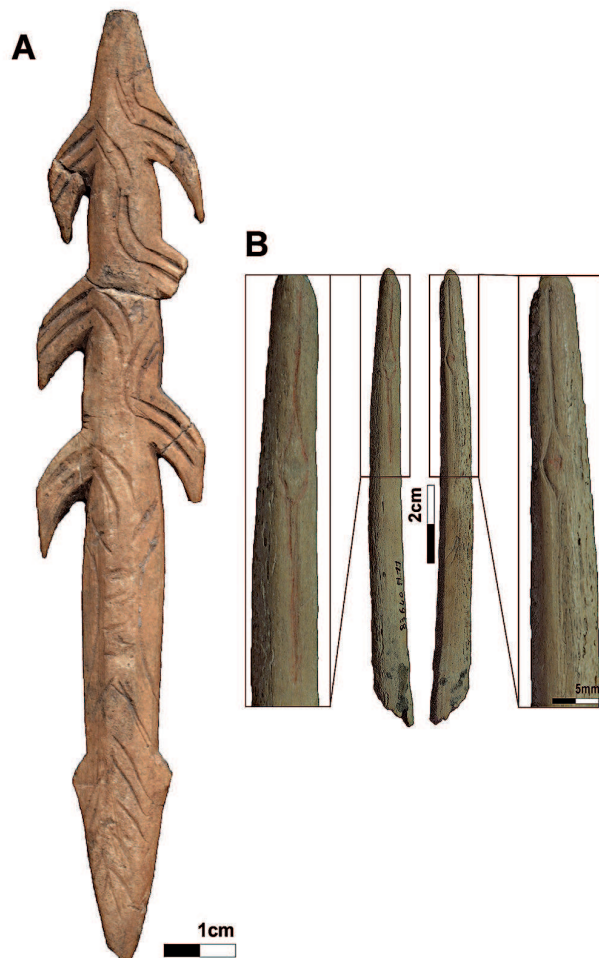


Fig. 3
Examples of
Magdalenian
decorated weaponry:
(A) Engraved
bilaterally barbed
point (Grotte
du Bois-Ragot),
(B) Engraved and
painted point
(La Vache).

As neither Africa nor Australia appears to have had an *extensive* osseous projectile technology, lithic hunting weaponry was widely employed and, in the case of Australia, wooden projectile technology was also manufactured throughout the Pleistocene.

Hunting weaponry, like body ornamentation, would have been highly visible to individuals who were both intimately familiar with the individual carrying the implement as well as those encountered out on the landscape during subsistence activities. Consequently, it is argued that these artefacts would have played a role in boundary maintenance in much the same way as body ornamentation or rock art might have: that is, providing information about the owners identity, affiliations and status to those in 'the middle distance'.

APPEARANCE IN THE ARCHAEOLOGICAL RECORD

Symbolic material culture can enter the archaeological record through a variety of complex and often interrelated processes. A few of these processes are outlined below.

Increase/Decrease in Population Densities

Recent studies (Shennan 2000, 2001, Powell, Shennan, Thomas 2009, Kline, Boyd 2010) have demonstrated the link between population density, interaction and cultural innovation, where larger populations generate and retain greater cultural diversity than smaller, isolated populations.

In regards to the appearance of symbolic material culture in the archaeological record, these studies suggest that regions where population density was higher is more likely to have both a greater diversity of material culture as well as more abundant material culture. This effect not only helps to explain the differences in the symbolic material culture witnessed in Upper Palaeolithic (and other) Modern Humans, but may also at least partially address the differences between the Middle and Upper Palaeolithic archaeological records.

Middle Palaeolithic population densities are argued to have been smaller and more widely dispersed than those for the following Upper Palaeolithic periods (Bar-Yosef 2002, Dickson 1990, Hoffercker 2002, Shennan 2001, Stiner, Munro, Surovell et al. 1999, Stiner, Munro, Surovell 2000, Stringer, Gamble 1993, see Hayden 2012 for discussion). Similarly, it is argued that Modern Humans interacted with neighbouring and/or more distant groups more frequently than Neanderthals (Adler, Bar-Oz, Belfer-Cohen et al. 2006, Féblot-Augustins 1999, Gamble 1999). Furthermore, it is suggested that Modern Human groups were not only significantly larger than those of Neanderthals but also were compositionally quite different, with the former populations' increased life expectancy resulting in considerably more elders constituting part of the community. The presence of these older individuals in turn would have resulted in the potential adoption of more clearly defined sex related roles in the community (Kuhn, Stiner 2006). All of these demographic factors may have resulted in a different need and use of symbolic material culture (Gamble 1982).

Differences in population density, historical contingency, environment and economy then all shape human societies in various ways giving rise to the immense cultural diversity documented throughout human history. These factors were likely to have been strong shaping factors in the cultural differences between populations such as Pleistocene Australians and European Aurignacians, for instance, who, while being Modern Humans with identical cognitive abilities and near contemporaries, as founding populations on new continents possessed markedly different archaeological signatures of symbolic expression.

Loss, Discard and Curation Practices

The archaeological record is primarily made up of prehistoric rubbish. The items people made, used and discarded – either intentionally or unintentionally – have entered the archaeological record over thousands of years, creating the record which we have to study.

Items may enter the archaeological record unintentionally. Beads sewn on to clothing or ornamentalations worn on the body may become loose during everyday activities and be subsequently lost. Similarly, decorated weapons may become lost to their owners if, during hunting, they are thrown and cannot be retrieved. These items, if they are not then destroyed by natural processes while lying on the surface, then enter the archaeological record scattered around the landscape. Similarly, items that are cached – that is, kept in a special place until they are needed – can enter the archaeological record through being cached at the end of a season and then forgotten, never to be retrieved.

Other items will be intentionally discarded. Beads, and other items, which may hold social information and value, often represent wealth in a community and may be inherited, distributed, exchanged, intentionally destroyed or abandoned in a ritualised context such as a burial (Vanhaeren, d’Errico 2005). Similarly, tools and weapons will be discarded when they become unusable – this usability being determined by both cultural and physical factors.

Other items may be ritually destroyed, such as weapons being sacrificially thrown into a fire or otherwise disposed of, as has been documented ethnographically (e.g. Marshall 1996, Osgood 1940), and will subsequently be irretrievable to archaeologists.

Some symbolic material culture such as beads, decorated tools and/or weapons or small art objects – if not intentionally abandoned in caves or graves – may be curated for significant periods of time, often being handed down from generation to generation as discussed for mobiliary art above. These items only entered the archaeological record after a significant period of time since their initial manufacture and possibly were deposited at a place far from their origin.

TAPHONOMY

While significant advances in our understanding of the origins and development of symbolic behaviour have been made over the past 150 years, the impact that taphonomic processes have had on the archaeological records under study has not played as prominent a role in this research as perhaps it should.

The kinds of material culture used by archaeologists to identify symbolic behaviour in the archaeological record are dominated by organic based artefacts and features (burials, shell and bone beads, mobile art, decorated osseous weapons and tools etc.) as well as fragile features such as rock art. The use of so much organic based and archaeologically fragile evidence for identifying the first appearance of symbolic behaviour and its consequent development in the archaeological record indicates that taphonomy should be a key consideration when identifying and describing spatial and temporal patterns. However, this has not always been the case.

Several researchers have cautioned about the potential impact that taphonomy and sample size may have on our global interpretations of the early record of modern cognition – of which symbolic behaviour is a key part (e.g. Bednarik 1992, McBrearty, Brooks 2000, d’Errico 2003, Henshilwood, Marean 2003, Langley 2014, Wadley 2001). However, the majority have only discussed taphonomy in relation to specific artefacts or features central to the debate: for example, burials (McBrearty, Brooks 2000, d’Errico 2003), pigment (Bednarik 1992, McBrearty, Brooks 2000, d’Errico 2003), osseous technology (Bednarik 1992, d’Errico 2003, Henshilwood, Marean 2003), and the impact of sampling on faunal assemblages and their subsequent interpretation (Mellars 1989a, 1989b, Marean 1998, Marean, Kim 1998, Marean, Assefa 1999, McBrearty, Brooks 2000, Stiner 2002, d’Errico 2003, Henshilwood, Marean 2003), rather than the impact of these processes on the wider regional patterns of behaviour.

The potential impact that taphonomic factors may have on interpretations of symbolic behaviour in the archaeological record becomes no more apparent than when focusing on the Pleistocene archaeological record of Sahul (the combined Pleistocene low sea-level landmass of New Guinea and Australia).

Although sites are represented back to >50,000 years BP, evidence for symbolic behaviour first appears between 41,000 and 45,000 years BP. The use of pigment at sites such as Malakunanja II (ochre crayon) (Roberts, Jones, Spooner et al. 1994), Carpenter's Gap Rockshelter 1 (painted roof fall) (O'Connor 1995, O'Connor, Fankhauser 2001) and Lake Mungo (exotic ochre) (Allen 1972, Bowler 1998) are the first evidence for symbolic activities in Sahul. Pigment use provides the most abundant form of symbolic evidence throughout the Pleistocene. The early use of pigment at Lake Mungo (c.40,000 years BP) is potentially the most significant of these owing to its use in the WLH3 ritual extended burial and the transport of ochre over c.200km to the burial site (Bowler, Johnston, Olley et al. 2003). The world's earliest cremation is also found at this site, dating to c. 40,000 years BP, together demonstrating the ritual complexity of early populations in southeast Australia (Bowler, Jones, Allen et al. 1970, Bowler, Johnston, Olley et al. 2003).

The first evidence for rock art appears early in the record (42800±1850 OZD161; 38177 ANUA7626) at Carpenter's Gap Rockshelter 1 (O'Connor, Fankhauser 2001), and remains one of the main components of the symbolic record throughout the Pleistocene. The decoration and/or transfer of cultural meanings through this medium therefore contribute one of the oldest symbolic behaviours in Sahul, as well as the most frequently identified in the archaeological record (see Langley and Tàyon 2010 for an analysis of dated Australian rock art).

Burials at Lake Mungo (Allen 1972, Bowler 1998) and personal ornamentation from Riwii (pierced shell beads) (Balme 2000, Balme, Morse 2006), Mandu Mandu Creek (pierced shell beads) (Morse 1993, Balme, Morse 2006) and Buang Merabak (drilled shark's tooth) (Leavesley 2007) first appear between 36,000 – 40,000 years BP; however, these traits only contribute sporadically to the record of complex behaviour until the early Holocene. Evidence for mobiliary art or notational pieces (as mentioned above) appears on only four occasions, between c. 25,000 – 13,000 years BP. Finally, intentional cranial deformations along with possible cemeteries at Coobool Creek and Kow Swamp (Brown 1987, Pardoe 1993) appear in the last 5,000 years of the Pleistocene.

Langley et al. (2011) examined the temporal and spatial distribution of evidence for complex (including symbolic) behaviours in the Pleistocene Sahul archaeological record and through analysis of site type, assemblage composition, environmental factors, excavation size and excavation methods concluded that the patterning of evidence was not the result of "*chronological and geographic patterning*" (Habgood, Franklin 2008: 210), but rather heavily influenced by taphonomic factors in conjunction to an increasing population size throughout the Pleistocene.

In particular, this study identified a pattern of increasing number of both occupied sites and quantity of evidence present in the archaeological record. This pattern easily fits Surovell and Brantingham's (2007) and Surovell, Finley, Smith et al.'s (2009) taphonomic degradation curve, and a significant correlation between the number of sites occupied versus the quantity of evidence was found. Therefore, the abundance of evidence identified through time is tied to the sample size available.

The significant increase in the quantity of symbolic evidence in the last 5,000 years of the Pleistocene suggests that there was either an increased need for symbolic communication amongst the populations of Sahul, perhaps caused by increased population density or economic stress as a result of the LGM, or simply that these artefacts and features are better preserved in the terminal Pleistocene. A combination of these two scenarios probably represents the best interpretation of these data.

The pattern of increasing quantity and diversity of evidence for complex (including symbolic) behaviours identified in the archaeological record of Pleistocene Sahul is not entirely unique. Similar patterns have been identified in the archaeological records of both Middle Palaeolithic Neanderthals (Langley, Clarkson, Ulm 2008) and MSA Africa (McBrearty, Brooks 2000). If the pattern

present within Sahul's Pleistocene archaeological record is the result of both increasing population density and taphonomic factors, does this imply that the similar trends for Middle Palaeolithic Eurasia and MSA Africa was also the product of similar demographic and taphonomic factors? Further detailed investigation is required to further elucidate the impact taphonomic factors have had on the archaeological records from which we build our knowledge of the use and development of symbolic behaviour throughout human evolution.

CONCLUSION

Through focusing on four categories of artefacts – body ornamentation, hunting weaponry, mobility art, and rock art – this paper has outlined the possible ways in which these material culture items were used in the Late Pleistocene as well as highlighted the difficulties of investigating the development and use of symbolic material culture during this period. While it is in no way exhaustive in its discussion of the issues inherent in this line of research, it nonetheless provides an understanding of the complexities various anthropogenic and taphonomic processes have on the formation of the archaeological record of symbolic behaviour in later human evolution. Of all lines of research in human origins studies, the investigation of symbolic behaviour perhaps requires the most caution when building interpretations from the record which survives to be examined.

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